Cold Fusion in a Metal: A Restoring Seat in an Octahedral Cage

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Abstract

Cold fusion, in this theory, is the slow charged entrance of a like pair at room temperature. Its host along one axis is a cage of like ions. A regular octahedron of positive ions holds one further positive ion at its center and pushes an electron off that center, so the electrons of the metal are not a screen. Two protons on an axis of the same octahedron, of cage radius 2ℓ, have a separation-increasing component that is strictly negative at separation 2ℓ, for every ion charge at least one. Just outside the first charge node the component tends to $+\infty$: the partner's repulsion diverges, and the six ions stay off the node. Between the node and separation 2ℓ the component falls strictly, and it vanishes at exactly one separation. Toward the node the component is outward, and toward 2ℓ it is inward, so both sides drive the pair to that separation. The cage does not carry the pair through the node. The seat does not set the ratio of the two lengths, and it does not derive a rate, an excess of heat, or a choice among experiments.

1 The cage

Cold fusion is the slow charged entrance fixed by the nuclear companion [1]. It is the entrance of a like pair whose kinetic energy is that of room temperature. Below the rapidity threshold the nuclear term opposes the Coulomb term, and next to the threshold it exceeds every finite ratio of the two prefactors, so long as the electromagnetic factor stays positive. A free infall meets the threshold before the node. The host of the holding, along one axis, is the electromagnetic channel already fixed for charges [2]. No second coupling is introduced.

On the equal-scale locus the force that increases the separation of two charges is

$$F(r) = \frac{k q_1 q_2}{\gamma_s(r)\, r^2}, \qquad k = \frac{1}{4\pi\epsilon_0}, \tag{1}$$

with $\gamma_s(x) = \cosh x \cos x - \sinh x \sin x$ and $x = \ell/(2r)$. Like charges repel where $\gamma_s > 0$ and attract where $\gamma_s < 0$. The first node x_1 is the unique root of $\tan x = \coth x$ in $(\pi/4, \pi/2)$, at $r_1/\ell = 0.53330$. The outer well is $r > r_1$, where $\gamma_s > 0$.

A regular octahedron of positive ions holds a further positive ion at its center and pushes an electron off that center. The force at the center vanishes by the symmetry, and every displacement of the positive ion is restored, while every displacement of an electron is pushed further. Under the Coulomb law both rates are zero. That zero is Earnshaw's theorem, and the signed factor removes it. A metal in the outer well is a held arrangement of ions, and the electrons have no seat among them. Screening is not introduced. Each ion remains a point charge.

The nearest positive ions about an octahedral hole of the face-centered cubic arrangement form this octahedron. Palladium and nickel are of that arrangement. The name records the geometry of the hole. Fleischmann and Pons placed deuterium in palladium and asked whether a nuclear entrance occurred there [3]. The seat below does not decide what that arrangement produced.

2 The axial component

The centered ion is one charge, restored in every direction. Two protons on an axis are a pair, and the component computed here is the component along that axis. Place ions of charge Ze at $(\pm a, 0, 0)$, $(0, \pm a, 0)$, and $(0, 0, \pm a)$, and protons of charge $+e$ at $(\pm t, 0, 0)$, with $0 < t < a$. Every ion–proton

separation and the proton–proton separation are taken off every node. The component that increases the separation is

$$F_{\parallel}(t) = -ZF(a-t) + ZF(a+t) + 4Z\frac{t}{R}F(R) + F(2t), \tag{2}$$

where $R = \sqrt{a^2+t^2}$ and F is (1) for two unit charges. The first term is the nearer axial ion. On the outer well it is a repulsion, and the direction from that ion toward the proton points inward, so the term decreases the separation. The second term is the farther axial ion, and it increases the separation. The third is the four equatorial ions: each repulsion has an outward axial projection t/R. The last term is the partner. On the outer well, $F_{\parallel}$ is a contest among repulsions.

A deuteron carries the same charge $+e$, so (2) is the axial component of a deuteron pair in the same cage. The coordination saddles of a metallic bond, at eight and at twelve neighbors, are saddles of a reduced electron content. They are not this interstitial seat.

3 Outward outside the node, inward at twice the length

Take the cage radius to be $a = 2\ell$, and the half-separation to be $t = \ell$. The proton separation is then 2ℓ. The nearer ion stands at ℓ, the farther at 3ℓ, each equatorial ion at $\sqrt{5}\,\ell$, and the partner at 2ℓ. Each of these distances lies in the outer well. For every ion charge $Z \geq 1$,

$$F_{\parallel}(\ell) < 0. \tag{3}$$

The nearer ion outweighs the farther ion, the four equatorial ions, and the partner together. Its phase is the largest, so $1/\gamma_s$ amplifies it more than the others. The cage contribution at this separation is itself inward, and a larger Z adds to that inward side. The partner does not depend on Z.

As the separation decreases to the first charge node, $2t \downarrow r_1$, the partner's phase increases to x_1 and $F(2t)$ tends to $+\infty$. The six ions remain off the node: the nearest approaches $2\ell - r_1/2$, still outside r_1, and the other five are farther. Their contribution tends to a finite value, so

$$F_{\parallel}(t) \to +\infty \qquad \text{as} \qquad 2t \downarrow r_1. \tag{4}$$

The component is outward in a punctured right neighborhood of the node and inward at separation 2ℓ. It is continuous on the intervening separations, all of which stay in the outer well, and on those separations it falls strictly. The zero is therefore unique:

$$r_1 < s < 2\ell, \qquad F_{\parallel}(s/2) = 0. \tag{5}$$

On the side toward the node the component is outward, and on the side toward separation 2ℓ it is inward. Both sides drive the pair to that separation. The cage does not carry the pair through the node.

The divergence that stops the approach sits at the finite radius r_1, where γ_s vanishes. Under the Coulomb law the partner term diverges only at coincidence, and the three-direction sum on a single charge is zero, so the cage does not hold one proton at its center. The signed factor holds one positive ion at the center, and it stops a pair at a finite node.

4 What the seat leaves open

The seat is that one separation, and it keeps the pair in $(r_1, 2\ell)$. The relative rapidity of a free infall is the rapidity of a separation that is falling toward the node, and the seat keeps the separation off that fall. Room temperature remains the class of the nuclear companion: a kinetic energy at most 10^{-10} of the rest energy. The nuclear comparison for the pair remains that entrance. The electromagnetic factor is positive at the nuclear node, and the rapidity is below the threshold.

The seat is a statement in the electromagnetic length. The nuclear entrance is a statement in the nuclear length. The cage does not set their ratio. If the charge node lies outside the nuclear node, the balance (5) stands outside an infinite electromagnetic wall, and the classical path on the axis does not reach the nuclear skin. The seat does not derive a rate, an excess of heat, or a choice among the experiments that have asked whether cold fusion occurs in a metal.

References

[1] Anonymous, "The Strong Force as a Repulsive Layer of Proximity Gravity: Proton Chaos, Magic Numbers, Slow Neutrons, and Cold Fusion," companion note.

[2] Anonymous, "Electron Orbits from the Signed Electrostatic Force: Covalent Bonds, Metallic Bonds, and Crystals," companion note.

[3] M. Fleischmann and S. Pons, "Electrochemically induced nuclear fusion of deuterium," *J. Electroanal. Chem.* **261**, 301 (1989).